Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ]

[18MT52]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

THEORY & TECHNOLOGY OF HEAT TREATMENT

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	a.	Define heat treatment. Mention heat treatment process variables and write down the stages of heat treatment.	07
	b.	Explain Full annealing and Isothermal annealing. What is the major difference between them?	07
2.	a.	What is the significance of hardenability? Mention the available methods for determining the hardenability. Explain Jominy's end quench test method with a neat sketch.	08
	b.	Name some synthetic quenchants used in heat treatment process. How synthetic quenchants have better heat transfer characteristics?	06
3.	a.	Explain progressive flame hardening process with neat sketch. Which microstructure is obtained in steel after flame hardening process?	07
	b.	Explain laser hardening process and also mention the advantages of laser heat treatment.	07
		OR	
4.	a.	Mention the methods available for measuring case depth which is surface hardened. Explain any one method.	06
	b.	Explain vacuum carburizing process. What are the factors responsible for saving of energy in vacuum carburizing process?	08
5.	a.	What are the applications of stainless steels? Classify stainless steel. What is sensitization? How to avoid sensitization?	08
	b.	Differentiate between White Cast Iron and Spheroidal Graphite Iron.	06
		OR	
6.	a.	With a neat diagram, explain the heat treatment process in Aluminum alloys.	07
	b.	What is the purpose of heat treatment in Titanium alloys? Mention the types of heat treatment process adopted for Titanium alloys. Explain any one.	07
7.	a.	Express the relationship between austenite grain size and grain size index. Explain how decomposition of austenite takes place during heating.	06
	b.	How CCT diagrams are different from TTT curves? Draw CCT diagrams for eutectoid steels with an explanation.	08
		OR	
8.	a.	Briefly describe the kinetics of formation of austenite. What is the effect of temperature on completion of transformation of pearlite?	08
	b.	How spinodal decomposition occurs? What is miscibility gap?	06